

A Subsymmetric Envelope Obstruction for Causey's Question 6.1

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Status

This packet records a partial result related to R. M. Causey's paper *Three and a half asymptotic properties* [1]. Causey's Question 6.1 asks whether the classes $\mathfrak{T}_{\xi,p}$ and $\mathfrak{A}_{\xi,p}$ are distinct for all ordinals ξ and all $1 < p \leq \infty$. The result below does not answer the full question. It rules out a natural class of candidate envelopes for the remaining uncountable-cofinality case.

Source Question

Causey proves the strict inclusion $\mathfrak{T}_{\xi,p} \subsetneq \mathfrak{A}_{\xi,p}$ for successor ordinals and for limit ordinals of countable cofinality. The paper then asks:

Remark 6.14. The class of limit ordinals with countable cofinality is a proper class. Therefore Theorems 6.2 and 6.3 produce results for uncountable ordinals. For example, they yield the distinctness of $\mathfrak{T}_{\omega_1\omega,p}$ and $\mathfrak{A}_{\omega_1\omega,p}$, since $\omega_1\omega = \lim_n \omega_1 n$. Here ω_1 denotes the first uncountable ordinal.

Question 6.1. Are $\mathfrak{T}_{\xi,p}$ and $\mathfrak{A}_{\xi,p}$ distinct for all ordinals?

We last conclude with the promised modification of the example of Lindenstrauss. Let E be a Banach

The crop is from page 35 of the arXiv PDF [1].

Definitions Used

For an operator $A : X \rightarrow Y$ and an ordinal ξ , Causey defines a seminorm $\|\cdot\|_{\xi,A}$ on c_{00} by testing weakly null trees of order $\Gamma_{\xi,n}$ [1, Section 2]. For $1 < p < \infty$, an operator A lies in $\mathfrak{A}_{\xi,p}$ precisely when the formal identity

$$(c_{00}, \|\cdot\|_{\ell_p}) \longrightarrow (c_{00}, \|\cdot\|_{\xi,A})$$

is bounded. The analogous definition for $p = \infty$ uses the c_0 norm. Write $\alpha_{0,p}(I_E)$ for this operator norm when $A = I_E$ and $\xi = 0$.

Causey's countable-cofinality witness uses a countable T_q^* envelope. A tempting uncountable-cofinality strategy would be to replace T_q^* by a long symmetric or subsymmetric envelope whose countable coordinate subsequences still fail ℓ_p domination, while retaining the $\mathfrak{A}_{0,p}$ estimate needed in Causey's envelope lemma [1, Lemma 6.8 and Corollary 6.9]. The theorem below shows that this cannot happen under subsymmetry.

Idea of the Proof

The $\|\cdot\|_{0,E}$ seminorm sees ordinary weakly null finite arrays. In a space with a shrinking coordinate basis, the unit vectors form a weakly null tail net. If the basis is subsymmetric, every far-out increasing coordinate branch is equivalent to the initial segment of the basis. Therefore the $\|\cdot\|_{0,E}$ test already contains the original basis norm. If $I_E \in \mathfrak{A}_{0,p}$, that seminorm is bounded by the ℓ_p norm (or by the c_0 norm for $p = \infty$), so the basis itself must be ℓ_p -dominated (respectively c_0 -dominated).

The Partial Result

Theorem 1 (Subsymmetric obstruction). *Let E be a Banach space with a normalized shrinking K -subsymmetric basis $(e_j)_{j=1}^\infty$. Let $1 < p < \infty$. If $\alpha_{0,p}(I_E) < \infty$, then for every n and every scalar family $(a_i)_{i=1}^n$,*

$$\left\| \sum_{i=1}^n a_i e_i \right\|_E \leq K \alpha_{0,p}(I_E) \left(\sum_{i=1}^n |a_i|^p \right)^{1/p}.$$

If $p = \infty$ and $\alpha_{0,\infty}(I_E) < \infty$, then

$$\left\| \sum_{i=1}^n a_i e_i \right\|_E \leq K \alpha_{0,\infty}(I_E) \max_{1 \leq i \leq n} |a_i|.$$

Proof. Fix n and scalars $(a_i)_{i=1}^n$. Since the basis is shrinking, the unit vector sequence is weakly null. More concretely, for every weak neighborhood U of 0 and every $m_0 \in \mathbb{N}$, there is $m > m_0$ such that $e_m \in U$.

Build a weakly null collection on the height- n tree $\Gamma_{0,n}.D_E$ as follows. Along each node, after the previous chosen indices $m_1 < \dots < m_{i-1}$ have been fixed, choose $m_i > m_{i-1}$ with e_{m_i} in the weak neighborhood attached to that node, and put the tree vector equal to e_{m_i} . Thus every maximal branch produces an increasing subsequence $(e_{m_i})_{i=1}^n$ of the basis.

For $\xi = 0$, the weights in the tree norm put one vector on each of the n successive levels. Hence the branch sums tested in $\|\sum_{i=1}^n a_i e_i\|_{0,I_E}$ include

$$\sum_{i=1}^n a_i e_{m_i}$$

over increasing branches. By K -subsymmetry,

$$\left\| \sum_{i=1}^n a_i e_{m_i} \right\|_E \geq K^{-1} \left\| \sum_{i=1}^n a_i e_i \right\|_E$$

for every branch. Taking the infimum over maximal branches and then the supremum over weakly null collections gives

$$K^{-1} \left\| \sum_{i=1}^n a_i e_i \right\|_E \leq \left\| \sum_{i=1}^n a_i e_i \right\|_{0,I_E}.$$

If $1 < p < \infty$ and $\alpha_{0,p}(I_E) < \infty$, the definition of $\alpha_{0,p}$ gives

$$\left\| \sum_{i=1}^n a_i e_i \right\|_{0,I_E} \leq \alpha_{0,p}(I_E) \left(\sum_{i=1}^n |a_i|^p \right)^{1/p}.$$

Combining the last two inequalities proves the first assertion. The $p = \infty$ case is identical, replacing the ℓ_p^n norm by $\max_i |a_i|$. $\square$

Corollary 1 (Unordered symmetric version). *Let E have a normalized shrinking K -symmetric unconditional coordinate basis $(e_\gamma)_{\gamma \in \Gamma}$ indexed by an infinite set Γ . If $\alpha_{0,p}(I_E) < \infty$, then every countable coordinate subsequence is $K\alpha_{0,p}(I_E)$ -dominated by the ℓ_p basis for $1 < p < \infty$, and by the c_0 basis for $p = \infty$.*

Proof. The same proof uses the weakly null coordinate net obtained by ordering finite subsets of Γ by inclusion. Shrinking implies that outside a large enough finite set all coordinate vectors lie in any prescribed weak neighborhood of 0. Along each finite tree branch choose distinct coordinates. Symmetry makes the norm of every branch equivalent, with constant K , to any prescribed countable coordinate model. The same $\|\cdot\|_{0,I_E}$ estimate gives the domination. $\square$

Consequence for the Open Problem Route

The previous attempt on this target isolated a possible positive route for uncountable-cofinality limits: build an uncountable envelope E_Γ with

$$\alpha_{0,p}(I_{E_\Gamma}) < \infty$$

but such that every increasing countable coordinate subsequence fails ℓ_p domination (or fails c_0 domination when $p = \infty$). Such an envelope, combined with Causey’s envelope lemma and ravenous-tree lemma, would be a plausible replacement for the countable T_q^* construction.

The theorem and corollary show that this strategy cannot be implemented with a shrinking symmetric or subsymmetric coordinate envelope. In those cases, the required $\mathfrak{A}_{0,p}$ estimate already forces all countable coordinate subsequences to be ℓ_p -dominated (or c_0 -dominated), which is exactly incompatible with the intended non- $\mathfrak{T}_{\xi,p}$ witness.

Verification Notes

The proof is purely formal from Causey’s definition of $\|\cdot\|_{0,A}$ and the standard weak-null tail property of shrinking bases. No computational checker was used. The main point for human review is the use of the $\xi = 0$ tree norm: for $\Gamma_{0,n}$, the tested branches are ordinary finite weakly null arrays with one selected vector per level, so the coordinate branch constructed in the proof is a legitimate lower bound for $\|\cdot\|_{0,I_E}$.

Local duplicate checks found no promoted packet for this obstruction. A bounded web search on June 18, 2026 for the exact Question 6.1 wording, the paper title together with “uncountable cofinality”, and the symbols $\mathfrak{T}_{\xi,p}$ and $\mathfrak{A}_{\xi,p}$ found the source arXiv page but no later exact closure of Causey’s question. This packet claims only the route obstruction above, not a full solution of Question 6.1.

Human Review Recommendation

Review as a likely valid partial result. The most important checks are:

1. verify that the coordinate tail net from a shrinking unordered basis is weakly null in the sense used by Causey’s tree definition;

2. verify that the $\xi = 0$ tree weights reduce to the stated finite-array branch sums;
3. confirm that the consequence is not overstated as a full answer to Question 6.1.

References

- [1] R. M. Causey, *Three and a half asymptotic properties*, arXiv:1805.02747, 2018. <https://arxiv.org/abs/1805.02747>